Question Paper

Exam Date & Time: 19-Nov-2024 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

FIFTH SEMESTER B.TECH. (ELECTRONICS & COMMUNICATION ENGINEERING) EXAMINATIONS - NOVEMBER 2024
SUBJECT: HUM 3021: ENGINEERING ECONOMICS AND FINANCIAL MANAGEMENT

Marks: 50 **Duration: 180 mins.**

Answer all the questions.

- **Missing data may be suitably assumed.**
- **Interest factor table is provided for others use formula.**
- **Formula book is attached with the paper**

1A) A loan of ₹15,000 requires monthly payments of ₹477 over a 36-month period of time. These payments include both principal and interest. (4)
i) What is the nominal interest rate for this loan?
ii) What is the effective interest rate per year?
iii) Determine the amount of unpaid loan principal after 20 months.

1B) Tetrix is planning to expand its production operation. It has identified three different technologies for meeting the goal. The initial outlay and annual revenues with respect to each of the technologies are summarized in the Table below. Suggest the best technology that is to be implemented based on the present worth method of comparison assuming an interest rate of 15%, compounded annually. (3)

	Initial Outlay (₹)	Annual Revenue (₹)	Life (years)
Technology 1	12,00,000	4,00,000	10
Technology 2	20,00,000	6,00,000	10
Technology 3	18,00,000	5,00,000	10

15%

N	F/P	P/F	A/P	P/A	A/F	F/A	P/G	A/G
1	1.150	.8696	1.1500	.870	1.0000	1.000	0	0
2	1.323	.7561	.6151	1.626	.4651	2.150	.756	.465
3	1.521	.6575	.4380	2.283	.2880	3.473	2.071	.907
4	1.749	.5718	.3503	2.855	.2003	4.993	3.786	1.326
5	2.011	.4972	.2983	3.352	.1483	6.742	5.775	1.723
6	2.313	.4323	.2642	3.784	.1142	8.754	7.937	2.097
7	2.660	.3759	.2404	4.160	.0904	11.067	10.192	2.450
8	3.059	.3269	.2229	4.487	.0729	13.727	12.481	2.781
9	3.518	.2843	.2096	4.772	.0596	16.786	14.755	3.092
10	4.046	.2472	.1993	5.019	.0493	20.304	16.979	3.383
11	4.652	.2149	.1911	5.234	.0411	24.349	19.129	3.655
12	5.350	.1869	.1845	5.421	.0345	29.002	21.185	3.908

1C) Martin is planning for his retirement. He has 10 more years of service. He would like to deposit ₹8,500 at the end of the first year and thereafter he wishes to deposit the amount with an annual decrease of ₹500 for the next 9 years with an interest rate of 15%. Find the total amount accumulated at the end of the 10th year. (3)

15%

N	F/P	P/F	A/P	P/A	A/F	F/A	P/G	A/G
1	1.150	.8696	1.1500	.870	1.0000	1.000	0	0
2	1.323	.7561	.6151	1.626	.4651	2.150	.756	.465
3	1.521	.6575	.4380	2.283	.2880	3.473	2.071	.907
4	1.749	.5718	.3503	2.855	.2003	4.993	3.786	1.326
5	2.011	.4972	.2983	3.352	.1483	6.742	5.775	1.723
6	2.313	.4323	.2642	3.784	.1142	8.754	7.937	2.097
7	2.660	.3759	.2404	4.160	.0904	11.067	10.192	2.450
8	3.059	.3269	.2229	4.487	.0729	13.727	12.481	2.781
9	3.518	.2843	.2096	4.772	.0596	16.786	14.755	3.092
10	4.046	.2472	.1993	5.019	.0493	20.304	16.979	3.383
11	4.652	.2149	.1911	5.234	.0411	24.349	19.129	3.655
12	5.350	.1869	.1845	5.421	.0345	29.002	21.185	3.908
13	6.153	.1625	.1791	5.583	.0291	34.352	23.135	4.144
14	7.076	.1413	.1747	5.724	.0247	40.505	24.972	4.362
15	8.137	.1229	.1710	5.847	.0210	47.580	26.693	4.565

2A) Assume a company purchases a new Hydrogen fuel cell electric vehicle (FCEV) to incorporate into its sustainable fleet. The vehicle was acquired at an initial cost of Rs. 18,50,000. Given that the company anticipates using the FCEV for an economic life span of 5 years with a salvage value of ₹4,50,000, they intend to depreciate the vehicle over this period. Using the straight-line method of depreciation, calculate the annual depreciation expense and prepare a 5-year depreciation schedule for the FCEV. (4)

2B) The Smith Corporation, a mail-order firm, wants to install an automatic mailing system to handle product announcements and invoices. The firm has a choice between two different types of machines. The two machines are designed differently but have identical capacities and do the same job. The semiautomatic machine (model A) costs Rs 1,30,000 and will lasts for three years, while the fully automatic machine (model B) will cost Rs 1,56,000 and lasts for four years. The expected cash flows for the two machines, including maintenance, and salvage value are as follows: (3)

In Rs	Model A	Model B
Annual operating costs	60,000	45,000
Salvage value	22,000	18,000

Using a study period of five years with an interest rate of 10 %, which model must be installed? Assume market value equal to Rs 25,000 for a two-year-old semiautomatic machine and Rs 35,000 for a one-year-old fully automatic machine.

10%

n	F/P	P/F	A/F	A/P	F/A	P/A	A/G	P/G
1	1.100	.9091	1.0000	1.1000	1.000	0.909	0	0
2	1.210	.8264	.4762	.5762	2.100	1.736	0.476	0.826
3	1.331	.7513	.3021	.4021	3.310	2.487	0.937	2.329
4	1.464	.6830	.2155	.3155	4.641	3.170	1.381	4.378
5	1.611	.6209	.1638	.2638	6.105	3.791	1.810	6.862
6	1.772	.5645	.1296	.2296	7.716	4.355	2.224	9.684
7	1.949	.5132	.1054	.2054	9.487	4.868	2.622	12.763

2C) A town is planning to provide for its' water needs. Two types of alternatives are being considered. The first is to construct a full capacity tunnel now for $250,000. The second proposes building a half capacity tunnel now at $100,000, which is adequate for 20 years, and then building an additional half capacity tunnel at the same cost. The maintenance for the full capacity tunnel is $10,000 every 10 years and each half capacity tunnel is $6,000 every 10 years. The estimated additional costs in single half capacity tunnel (due to friction losses) are $1,000 every year. Based on capitalized cost and a 6% interest rate, which alternative should be selected? (3)

6%

n	F/P	P/F	A/F	A/P	F/A	P/A	A/G	P/G
1	1.060	.9434	1.0000	1.0600	1.000	0.943	0	0
2	1.124	.8900	.4854	.5454	2.060	1.833	0.485	0.890
3	1.191	.8396	.3141	.3741	3.184	2.673	0.961	2.569
4	1.262	.7921	.2286	.2886	4.375	3.465	1.427	4.945
5	1.338	.7473	.1774	.2374	5.637	4.212	1.884	7.934
6	1.419	.7050	.1434	.2034	6.975	4.917	2.330	11.459
7	1.504	.6651	.1191	.1791	8.394	5.582	2.768	15.450
8	1.594	.6274	.1010	.1610	9.897	6.210	3.195	19.841
9	1.689	.5919	.0870	.1470	11.491	6.802	3.613	24.577
10	1.791	.5584	.0759	.1359	13.181	7.360	4.022	29.602
11	1.898	.5268	.0668	.1268	14.972	7.887	4.421	34.870
12	2.012	.4970	.0593	.1193	16.870	8.384	4.811	40.337
13	2.133	.4688	.0530	.1130	18.882	8.853	5.192	45.963
14	2.261	.4423	.0476	.1076	21.015	9.295	5.564	51.713
15	2.397	.4173	.0430	.1030	23.276	9.712	5.926	57.554
16	2.540	.3936	.0390	.0990	25.672	10.106	6.279	63.459
17	2.693	.3714	.0354	.0954	28.213	10.477	6.624	69.401
18	2.854	.3503	.0324	.0924	30.906	10.828	6.960	75.357
19	3.026	.3305	.0296	.0896	33.760	11.158	7.287	81.306
20	3.207	.3118	.0272	.0872	36.786	11.470	7.605	87.230

3A) A automotive company is deciding between two alternatives, Alternative A is robotic automation system and Alternative B flexible manufacturing system as a part of its strategy to diversify its operation in a new location. The company sets its Minimum Acceptable Rate of Return (MARR) at 12% per year. (4)

	Alternative A	Alternative B
Initial Cost	$1,20,000	$90,000
Salvage Value (at end of 8th year)	$20,000	$10,000
Maintenance Cost	$6000 per year	$12 per hour
Operating Cost	$50 per hour	$40 per hour
Wages for operator	$10 per hour	$30 per hour
Estimated useful life	8 years	8 years

a) Calculate the number of hours of operation per year (H) at which the total costs for both alternatives are equal (break-even point).

b) If the company expects an annual operation of 1000 hours which alternative you would suggest the company to buy.

12%

n	F/P	P/F	A/F	A/P	F/A	P/A	A/G	P/G
1	1.120	.8929	1.0000	1.1200	1.000	0.893	0	0
2	1.254	.7972	.4717	.5917	2.120	1.690	0.472	0.797
3	1.405	.7118	.2963	.4163	3.374	2.402	0.925	2.221
4	1.574	.6355	.2092	.3292	4.779	3.037	1.359	4.127
5	1.762	.5674	.1574	.2774	6.353	3.605	1.775	6.397
6	1.974	.5066	.1232	.2432	8.115	4.111	2.172	8.930
7	2.211	.4523	.0991	.2191	10.089	4.564	2.551	11.644
8	2.476	.4039	.0813	.2013	12.300	4.968	2.913	14.471
9	2.773	.3606	.0677	.1877	14.776	5.328	3.257	17.356
10	3.106	.3220	.0570	.1770	17.549	5.650	3.585	20.254
11	3.479	.2875	.0484	.1684	20.655	5.938	3.895	23.129
12	3.896	.2567	.0414	.1614	24.133	6.194	4.190	25.952

3B) In an automotive parts plant, an engineering team is analysing an improvement project to increase the productivity of a flexible manufacturing centre. The estimated net cash flows for the three feasible alternatives being compared are shown in Table below. The analysis period is 6 years, and MARR for capital investments at the plant is 15% per year. Using Incremental Rate of Return method, which alternative should be selected? (4)

End of Period	A	B	C
0	- 640	- 680	- 755
1	+262	- 40	+205
2	+290	+392	+406
3	+302	+380	+400
4	+310	+380	+390
5	+310	+380	+390
6	+260	+380	+324
IRR	*38%*	*30%*	*37%*
In thousands of Indian Rupees (₹ '000)			

15%

N	F/P	P/F	A/P	P/A	A/F	F/A	P/G	A/G
1	1.150	.8696	1.1500	.870	1.0000	1.000	0	0
2	1.323	.7561	.6151	1.626	.4651	2.150	.756	.465
3	1.521	.6575	.4380	2.283	.2880	3.473	2.071	.907
4	1.749	.5718	.3503	2.855	.2003	4.993	3.786	1.326
5	2.011	.4972	.2983	3.352	.1483	6.742	5.775	1.723
6	2.313	.4323	.2642	3.784	.1142	8.754	7.937	2.097
7	2.660	.3759	.2404	4.160	.0904	11.067	10.192	2.450
8	3.059	.3269	.2229	4.487	.0729	13.727	12.481	2.781
9	3.518	.2843	.2096	4.772	.0596	16.786	14.755	3.092
10	4.046	.2472	.1993	5.019	.0493	20.304	16.979	3.383
11	4.652	.2149	.1911	5.234	.0411	24.349	19.129	3.655
12	5.350	.1869	.1845	5.421	.0345	29.002	21.185	3.908

3C) Ramu, a salesman, needs a new car for use in his business. He expects that he will be promoted to a supervisory job at the end of third year and so his concern is to have a car for the three years he expects to be "on the road". The company will reimburse the salesman each month for the fuel cost and maintenance. Ramu has decided to drive a low-priced automobile. However, he finds that there are two options to obtain the car. In either case, the fuel cost and maintenance cost are borne by the company. (2)

a) Purchase for cash at Rs. 3,90,000 and the car can be sold at Rs. 1,20,000 at the end of third year.

b) Lease a car. The monthly charge is Rs. 10,500 on a 36-month lease payable at the end of each month. At the end of the three-year period, the car is returned to the leasing company.

Using a MARR of 12% compounded monthly, which alternative should Ramu select? Apply annual worth method for analysis.

4A) A manufacturing company, XYZ Corp, has been using an old piece of machinery, Machine A, for several years. Recently, it has been experiencing frequent breakdowns, leading to increased repair costs and downtime. The management is considering whether to replace Machine A with a newer model, Machine B, which promises improved efficiency and lower operating costs. The details of both the machines are given below: (4)

Current Machine (A):	**New Machine (B):**
Age: 10 years	Initial Cost: $150,000
Remaining Useful Life: 5 years	Annual Operating Costs: $30,000
Annual Repair Costs: $10,000	Useful Life: 10 years
Annual Operating Costs: $50,000	Salvage Value: $20,000
Salvage Value: $5,000	
Note the current market value of Machine A is $50,000	

Using the appropriate replacement analysis approach with a 10% MARR, evaluate whether XYZ Corp should replace its old machine or not.

10% Compound Interest Factors 10%

n	Single Payment: Compound Amount Factor Find F Given P F/P	Single Payment: Present Worth Factor Find P Given F P/F	Uniform Payment Series: Sinking Fund Factor Find A Given F A/F	Uniform Payment Series: Capital Recovery Factor Find A Given P A/P	Uniform Payment Series: Compound Amount Factor Find F Given A F/A	Uniform Payment Series: Present Worth Factor Find P Given A P/A	Arithmetic Gradient: Gradient Uniform Series Find A Given G A/G	Arithmetic Gradient: Gradient Present Worth Find P Given G P/G	n
1	1.100	.9091	1.0000	1.1000	1.000	0.909	0	0	1
2	1.210	.8264	.4762	.5762	2.100	1.736	0.476	0.826	2
3	1.331	.7513	.3021	.4021	3.310	2.487	0.937	2.329	3
4	1.464	.6830	.2155	.3155	4.641	3.170	1.381	4.378	4
5	1.611	.6209	.1638	.2638	6.105	3.791	1.810	6.862	5
6	1.772	.5645	.1296	.2296	7.716	4.355	2.224	9.684	6
7	1.949	.5132	.1054	.2054	9.487	4.868	2.622	12.763	7
8	2.144	.4665	.0874	.1874	11.436	5.335	3.004	16.029	8
9	2.358	.4241	.0736	.1736	13.579	5.759	3.372	19.421	9
10	2.594	.3855	.0627	.1627	15.937	6.145	3.725	22.891	10
11	2.853	.3505	.0540	.1540	18.531	6.495	4.064	26.396	11
12	3.138	.3186	.0468	.1468	21.384	6.814	4.388	29.901	12
13	3.452	.2897	.0408	.1408	24.523	7.103	4.699	33.377	13
14	3.797	.2633	.0357	.1357	27.975	7.367	4.996	36.801	14
15	4.177	.2394	.0315	.1315	31.772	7.606	5.279	40.152	15

10% Compound Interest Factors 10%

n	Single Payment: Compound Amount Factor Find F Given P F/P	Single Payment: Present Worth Factor Find P Given F P/F	Uniform Payment Series: Sinking Fund Factor Find A Given F A/F	Uniform Payment Series: Capital Recovery Factor Find A Given P A/P	Uniform Payment Series: Compound Amount Factor Find F Given A F/A	Uniform Payment Series: Present Worth Factor Find P Given A P/A	Arithmetic Gradient: Gradient Uniform Series Find A Given G A/G	Arithmetic Gradient: Gradient Present Worth Find P Given G P/G	n
16	4.595	.2176	.0278	.1278	35.950	7.824	5.549	43.416	16
17	5.054	.1978	.0247	.1247	40.545	8.022	5.807	46.582	17
18	5.560	.1799	.0219	.1219	45.599	8.201	6.053	49.640	18
19	6.116	.1635	.0195	.1195	51.159	8.365	6.286	52.583	19
20	6.728	.1486	.0175	.1175	57.275	8.514	6.508	55.407	20
21	7.400	.1351	.0156	.1156	64.003	8.649	6.719	58.110	21
22	8.140	.1228	.0140	.1140	71.403	8.772	6.919	60.689	22
23	8.954	.1117	.0126	.1126	79.543	8.883	7.108	63.146	23
24	9.850	.1015	.0113	.1113	88.497	8.985	7.288	65.481	24
25	10.835	.0923	.0102	.1102	98.347	9.077	7.458	67.696	25
26	11.918	.0839	.00916	.1092	109.182	9.161	7.619	69.794	26
27	13.110	.0763	.00826	.1083	121.100	9.237	7.770	71.777	27
28	14.421	.0693	.00745	.1075	134.210	9.307	7.914	73.650	28
29	15.863	.0630	.00673	.1067	148.631	9.370	8.049	75.415	29
30	17.449	.0573	.00608	.1061	164.494	9.427	8.176	77.077	30

4B) A company is considering a new project that requires an initial investment of $100,000. The project is expected to generate cash flows as given below over the next five years, which is the most likely scenario. (4)

Year	1	2	3	4	5
Cashflow	$30,000	$40,000	$50,000	$20,000	$10,000

However, the company anticipates that cashflows will be 20% lower than projected each year in the worst-case scenario.

Calculate the Net Present Values (NPVs) for both the expected and worst-case scenarios, using a discount rate of 10%. Then, discuss the viability of investments by considering these two scenarios.

10% Compound Interest Factors 10%

n	Single Payment: Compound Amount Factor Find F Given P F/P	Single Payment: Present Worth Factor Find P Given F P/F	Uniform Payment Series: Sinking Fund Factor Find A Given F A/F	Uniform Payment Series: Capital Recovery Factor Find A Given P A/P	Uniform Payment Series: Compound Amount Factor Find F Given A F/A	Uniform Payment Series: Present Worth Factor Find P Given A P/A	Arithmetic Gradient: Gradient Uniform Series Find A Given G A/G	Arithmetic Gradient: Gradient Present Worth Find P Given G P/G	n
1	1.100	.9091	1.0000	1.1000	1.000	0.909	0	0	1
2	1.210	.8264	.4762	.5762	2.100	1.736	0.476	0.826	2
3	1.331	.7513	.3021	.4021	3.310	2.487	0.937	2.329	3
4	1.464	.6830	.2155	.3155	4.641	3.170	1.381	4.378	4
5	1.611	.6209	.1638	.2638	6.105	3.791	1.810	6.862	5
6	1.772	.5645	.1296	.2296	7.716	4.355	2.224	9.684	6
7	1.949	.5132	.1054	.2054	9.487	4.868	2.622	12.763	7
8	2.144	.4665	.0874	.1874	11.436	5.335	3.004	16.029	8
9	2.358	.4241	.0736	.1736	13.579	5.759	3.372	19.421	9
10	2.594	.3855	.0627	.1627	15.937	6.145	3.725	22.891	10
11	2.853	.3505	.0540	.1540	18.531	6.495	4.064	26.396	11
12	3.138	.3186	.0468	.1468	21.384	6.814	4.388	29.901	12
13	3.452	.2897	.0408	.1408	24.523	7.103	4.699	33.377	13
14	3.797	.2633	.0357	.1357	27.975	7.367	4.996	36.801	14
15	4.177	.2394	.0315	.1315	31.772	7.606	5.279	40.152	15

10% Compound Interest Factors 10%

n	Single Payment: Compound Amount Factor Find F Given P F/P	Single Payment: Present Worth Factor Find P Given F P/F	Uniform Payment Series: Sinking Fund Factor Find A Given F A/F	Uniform Payment Series: Capital Recovery Factor Find A Given P A/P	Uniform Payment Series: Compound Amount Factor Find F Given A F/A	Uniform Payment Series: Present Worth Factor Find P Given A P/A	Arithmetic Gradient: Gradient Uniform Series Find A Given G A/G	Arithmetic Gradient: Gradient Present Worth Find P Given G P/G	n
16	4.595	.2176	.0278	.1278	35.950	7.824	5.549	43.416	16
17	5.054	.1978	.0247	.1247	40.545	8.022	5.807	46.582	17
18	5.560	.1799	.0219	.1219	45.599	8.201	6.053	49.640	18
19	6.116	.1635	.0195	.1195	51.159	8.365	6.286	52.583	19
20	6.728	.1486	.0175	.1175	57.275	8.514	6.508	55.407	20
21	7.400	.1351	.0156	.1156	64.003	8.649	6.719	58.110	21
22	8.140	.1228	.0140	.1140	71.403	8.772	6.919	60.689	22
23	8.954	.1117	.0126	.1126	79.543	8.883	7.108	63.146	23
24	9.850	.1015	.0113	.1113	88.497	8.985	7.288	65.481	24
25	10.835	.0923	.0102	.1102	98.347	9.077	7.458	67.696	25
26	11.918	.0839	.00916	.1092	109.182	9.161	7.619	69.794	26
27	13.110	.0763	.00826	.1083	121.100	9.237	7.770	71.777	27
28	14.421	.0693	.00745	.1075	134.210	9.307	7.914	73.650	28
29	15.863	.0630	.00673	.1067	148.631	9.370	8.049	75.415	29
30	17.449	.0573	.00608	.1061	164.494	9.427	8.176	77.077	30

4C) Briefly explain why a long-term creditor should be interested in liquidity ratios. (2)

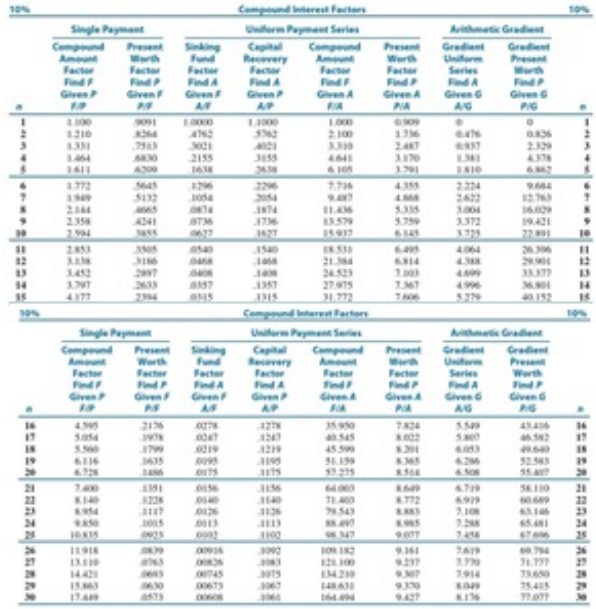
5A) A consulting engineering company is evaluating whether to purchase or lease one of these vehicles for its top management employee use. (3)

Purchase: Nissan's all-electric Leaf has a base price of ₹27,45,935 in India and is eligible for a ₹5,62,500 federal tax credit (benefit). If purchased it requires additional home charging station cost

set-up which costs ₹1,65,000 after a 50% tax credit. Expected resale value after 3 years for this car is ₹10,98,374 (which 40% of initial value).

Leasing: The leasing would have an annual cost of ₹3,48,900 (paid at the end of each year) with an initial lease charge of ₹1,82,500 (paid now).

Using an interest rate of 10% per year, perform an annual worth analysis to determine whether the company should purchase or lease the vehicle.



10% Compound Interest Factors 10%

n	Single Payment: Compound Amount Factor Find F Given P F/P	Single Payment: Present Worth Factor Find P Given F P/F	Uniform Payment Series: Sinking Fund Factor Find A Given F A/F	Uniform Payment Series: Capital Recovery Factor Find A Given P A/P	Uniform Payment Series: Compound Amount Factor Find F Given A F/A	Uniform Payment Series: Present Worth Factor Find P Given A P/A	Arithmetic Gradient: Gradient Uniform Series Find A Given G A/G	Arithmetic Gradient: Gradient Present Worth Find P Given G P/G	n
1	1.100	.9091	1.0000	1.1000	1.000	0.909	0	0	1
2	1.210	.8264	.4762	.5762	2.100	1.736	0.476	0.826	2
3	1.331	.7513	.3021	.4021	3.310	2.487	0.937	2.329	3
4	1.464	.6830	.2155	.3155	4.641	3.170	1.381	4.378	4
5	1.611	.6209	.1638	.2638	6.105	3.791	1.810	6.862	5
6	1.772	.5645	.1296	.2296	7.716	4.355	2.224	9.684	6
7	1.949	.5132	.1054	.2054	9.487	4.868	2.622	12.763	7
8	2.144	.4665	.0874	.1874	11.436	5.335	3.004	16.029	8
9	2.358	.4241	.0736	.1736	13.579	5.759	3.372	19.421	9
10	2.594	.3855	.0627	.1627	15.937	6.145	3.725	22.891	10
11	2.853	.3505	.0540	.1540	18.531	6.495	4.064	26.396	11
12	3.138	.3186	.0468	.1468	21.384	6.814	4.388	29.901	12
13	3.452	.2897	.0408	.1408	24.523	7.103	4.699	33.377	13
14	3.797	.2633	.0357	.1357	27.975	7.367	4.996	36.801	14
15	4.177	.2394	.0315	.1315	31.772	7.606	5.279	40.152	15

10% Compound Interest Factors 10%

n	Single Payment: Compound Amount Factor Find F Given P F/P	Single Payment: Present Worth Factor Find P Given F P/F	Uniform Payment Series: Sinking Fund Factor Find A Given F A/F	Uniform Payment Series: Capital Recovery Factor Find A Given P A/P	Uniform Payment Series: Compound Amount Factor Find F Given A F/A	Uniform Payment Series: Present Worth Factor Find P Given A P/A	Arithmetic Gradient: Gradient Uniform Series Find A Given G A/G	Arithmetic Gradient: Gradient Present Worth Find P Given G P/G	n
16	4.595	.2176	.0278	.1278	35.950	7.824	5.549	43.416	16
17	5.054	.1978	.0247	.1247	40.545	8.022	5.807	46.582	17
18	5.560	.1799	.0219	.1219	45.599	8.201	6.053	49.640	18
19	6.116	.1635	.0195	.1195	51.159	8.365	6.286	52.583	19
20	6.728	.1486	.0175	.1175	57.275	8.514	6.508	55.407	20
21	7.400	.1351	.0156	.1156	64.003	8.649	6.719	58.110	21
22	8.140	.1228	.0140	.1140	71.403	8.772	6.919	60.689	22
23	8.954	.1117	.0126	.1126	79.543	8.883	7.108	63.146	23
24	9.850	.1015	.0113	.1113	88.497	8.985	7.288	65.481	24
25	10.835	.0923	.0102	.1102	98.347	9.077	7.458	67.696	25
26	11.918	.0839	.00916	.1092	109.182	9.161	7.619	69.794	26
27	13.110	.0763	.00826	.1083	121.100	9.237	7.770	71.777	27
28	14.421	.0693	.00745	.1075	134.210	9.307	7.914	73.650	28
29	15.863	.0630	.00673	.1067	148.631	9.370	8.049	75.415	29
30	17.449	.0573	.00608	.1061	164.494	9.427	8.176	77.077	30

5B) Given the following financial data for ABC Corporation for the year ended December 31, 2023: (3)

- Total Revenue: $500,000
- Cost of Goods Sold (COGS): $300,000
- Net Income: $80,000
- Total Assets: $600,000
- Total Liabilities: $250,000
- Shareholders' Equity: $350,000

Calculate the following ratios:

a) Gross Profit Margin

b) Return on Assets (ROA)

c) Debt to Equity Ratio

5C) A new asset is available for $200,000. O&M costs are $20,000 each year for the first five years, $22,000 in year six, $25,000 in year seven and $28,000 in year eight. Salvage values are estimated to be $130,000 after year one and will decrease at the rate of 20% per year thereafter. At a MARR of 12% p.a., determine the economic service life of the asset. (4)

12% Compound Interest Factors 12%

n	Single Payment: Compound Amount Factor Find F Given P F/P	Single Payment: Present Worth Factor Find P Given F P/F	Uniform Payment Series: Sinking Fund Factor Find A Given F A/F	Uniform Payment Series: Capital Recovery Factor Find A Given P A/P	Uniform Payment Series: Compound Amount Factor Find F Given A F/A	Uniform Payment Series: Present Worth Factor Find P Given A P/A	Arithmetic Gradient: Gradient Uniform Series Find A Given G A/G	Arithmetic Gradient: Gradient Present Worth Find P Given G P/G	n
1	1.120	.8929	1.0000	1.1200	1.000	0.893	0	0	1
2	1.254	.7972	.4717	.5917	2.120	1.690	0.472	0.797	2
3	1.405	.7118	.2963	.4163	3.374	2.402	0.925	2.221	3
4	1.574	.6355	.2092	.3292	4.779	3.037	1.359	4.127	4
5	1.762	.5674	.1574	.2774	6.353	3.605	1.775	6.397	5
6	1.974	.5066	.1232	.2432	8.115	4.111	2.172	8.930	6
7	2.211	.4523	.0991	.2191	10.089	4.564	2.551	11.644	7
8	2.476	.4039	.0813	.2013	12.300	4.968	2.913	14.471	8
9	2.773	.3606	.0677	.1877	14.776	5.328	3.257	17.356	9
10	3.106	.3220	.0570	.1770	17.549	5.650	3.585	20.254	10
11	3.479	.2875	.0484	.1684	20.655	5.938	3.895	23.129	11
12	3.896	.2567	.0414	.1614	24.133	6.194	4.190	25.952	12
13	4.363	.2292	.0357	.1557	28.029	6.424	4.468	28.702	13
14	4.887	.2046	.0309	.1509	32.393	6.628	4.732	31.362	14
15	5.474	.1827	.0268	.1468	37.280	6.811	4.980	33.920	15

12% Compound Interest Factors 12%

n	Single Payment: Compound Amount Factor Find F Given P F/P	Single Payment: Present Worth Factor Find P Given F P/F	Uniform Payment Series: Sinking Fund Factor Find A Given F A/F	Uniform Payment Series: Capital Recovery Factor Find A Given P A/P	Uniform Payment Series: Compound Amount Factor Find F Given A F/A	Uniform Payment Series: Present Worth Factor Find P Given A P/A	Arithmetic Gradient: Gradient Uniform Series Find A Given G A/G	Arithmetic Gradient: Gradient Present Worth Find P Given G P/G	n
16	6.130	.1631	.0234	.1434	42.753	6.974	5.215	36.367	16
17	6.866	.1456	.0205	.1405	48.884	7.120	5.435	38.697	17
18	7.690	.1300	.0179	.1379	55.750	7.250	5.643	40.908	18
19	8.613	.1161	.0158	.1358	63.440	7.366	5.838	42.998	19
20	9.646	.1037	.0139	.1339	72.052	7.469	6.020	44.968	20
21	10.804	.0926	.0122	.1322	81.699	7.562	6.191	46.819	21
22	12.100	.0826	.0108	.1308	92.503	7.645	6.351	48.554	22
23	13.552	.0738	.00956	.1296	104.603	7.718	6.501	50.178	23
24	15.179	.0659	.00846	.1285	118.155	7.784	6.641	51.693	24
25	17.000	.0588	.00750	.1275	133.334	7.843	6.771	53.105	25
26	19.040	.0525	.00665	.1267	150.334	7.896	6.892	54.418	26
27	21.325	.0469	.00590	.1259	169.374	7.943	7.005	55.637	27
28	23.884	.0419	.00524	.1252	190.699	7.984	7.110	56.767	28
29	26.750	.0374	.00466	.1247	214.583	8.022	7.207	57.814	29
30	29.960	.0334	.00414	.1241	241.333	8.055	7.297	58.782	30

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